Lecture 3

Everything you didn't want to know about LM architecture and hyperparameters

CS336

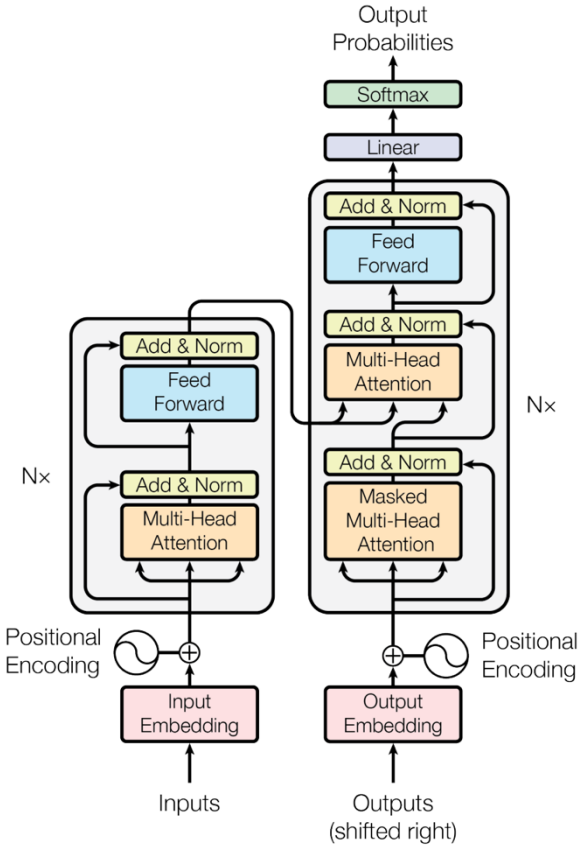
Tatsu H

Outline and goals

- Quick recap of a modern transformer (what you implement)
- What do most of the large LMs have in common?
- What are common variations to the architecture / training process?

Today's theme: the best way to learn is hands-on experience
the second best way is to try to learn from others' experience

Starting point: the ‘original’ transformer



Review: choices in the standard transformer

Position embedding: sines and cosines

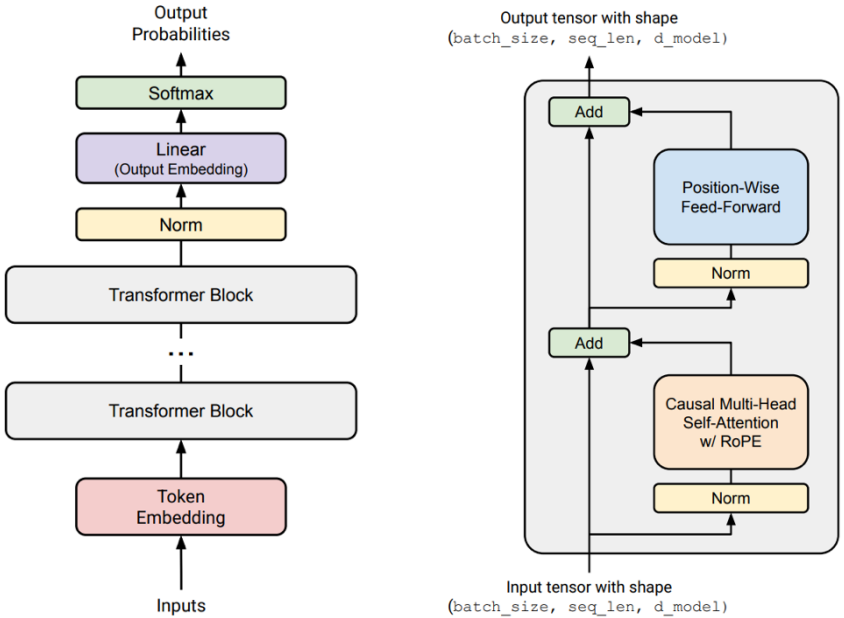
$$PE_{(pos,2i)} = sin(pos/10000^{2i/d_{\text{model}}})$$
$$PE_{(pos,2i+1)} = cos(pos/10000^{2i/d_{\text{model}}})$$

FFN: ReLU

$$\text{FFN}(x) = \max(0, xW_1 + b_1)W_2 + b_2$$

Norm type: post-norm, LayerNorm

What you implemented – simple, modern variant



Differences:

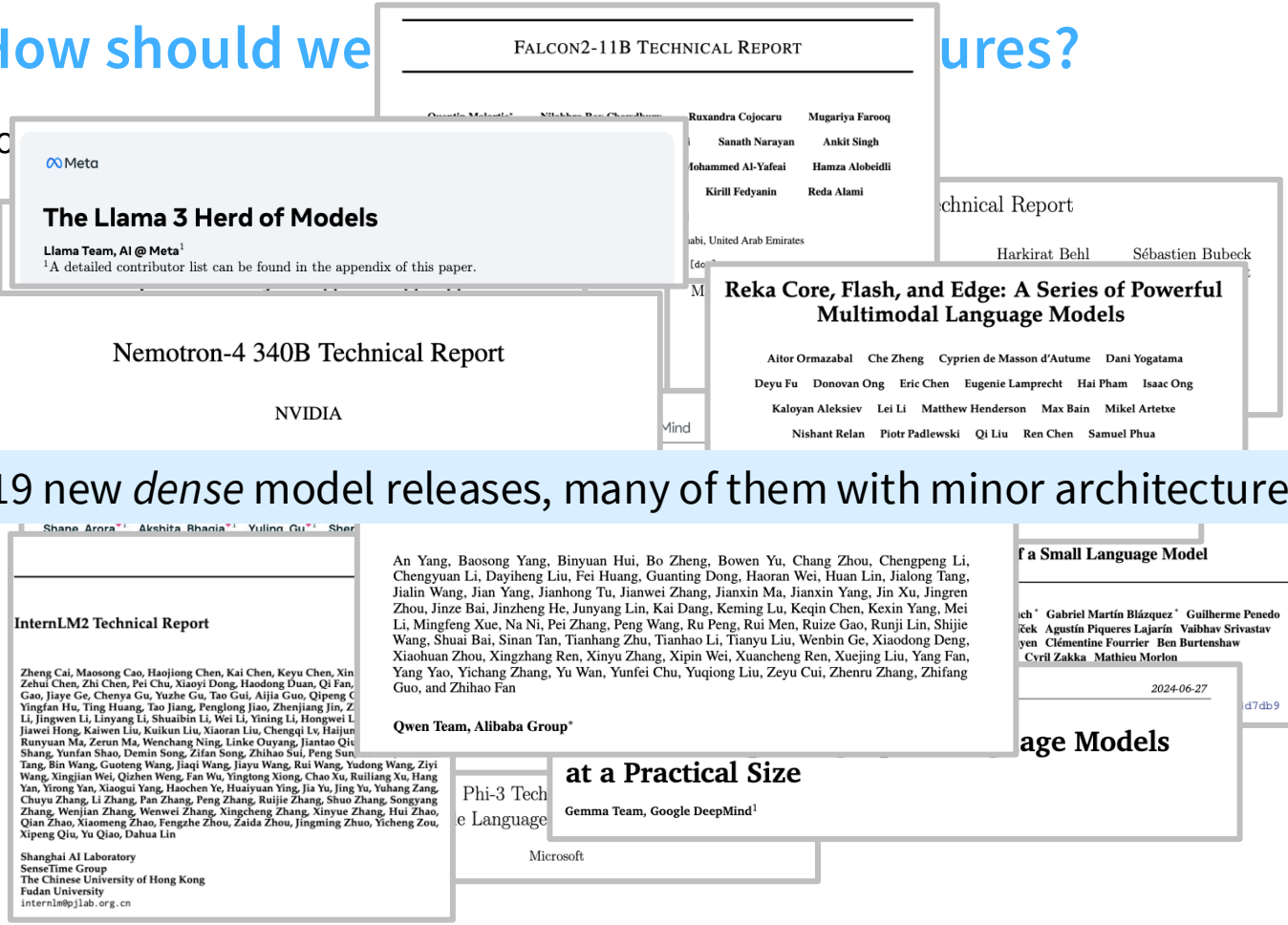
- **LayerNorm** is in front of the block
- **Rotary position embeddings (RoPE)**
- FF layers use **SwiGLU**, not ReLU
- Linear layers (and layernorm) have **no bias** (constant) terms

Why did we pick these?

What should you pick?

How should we [illegible] ures?

Lo[illegible]



FALCON2-11B TECHNICAL REPORT

Ruxandra Cojocaru Mugariya Farooq

Sanath Narayan Ankit Singh

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abi, United Arab Emirates

Meta

The Llama 3 Herd of Models

Llama Team, AI @ Meta[1]

[1]A detailed contributor list can be found in the appendix of this paper.

chnical Report

Harkirat Behl Sébastien Bubeck

Nemotron-4 340B Technical Report

NVIDIA

Reka Core, Flash, and Edge: A Series of Powerful Multimodal Language Models

Aitor Ormazabal Che Zheng Cyprien de Masson d'Autume Dani Yogatama

Deyu Fu Donovan Ong Eric Chen Eugenie Lamprecht Hai Pham Isaac Ong

Kaloyan Aleksiev Lei Li Matthew Henderson Max Bain Mikel Artetxe

Nishant Relan Piotr Padlewski Qi Liu Ren Chen Samuel Phua

Mind

Over 19 new *dense* model releases, many of them with minor architecture tweaks..

InternLM2 Technical Report

Zheng Cai, Maosong Cao, Haojiong Chen, Kai Chen, Keyu Chen, Xin
Zehui Chen, Zhi Chen, Pei Chu, Xiaoyi Dong, Haodong Duan, Qi Fan,
Gao, Jiaye Ge, Chenya Gu, Yuzhe Gu, Tao Gui, Aijia Guo, Qipeng G
Yingfan Hu, Ting Huang, Tao Jiang, Penglong Jiao, Zhenjiang Jin, Z
Li, Jingwen Li, Linyang Li, Shuaibin Li, Wei Li, Yining Li, Hongwei L
Jiawei Hong, Kaiwen Liu, Kuikun Liu, Xiaoran Liu, Chengqi Lv, Haijun
Runyuan Ma, Zerun Ma, Wenchang Ning, Linke Ouyang, Jiantao Qiu
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Tang, Bin Wang, Guoteng Wang, Jiaqi Wang, Jiayu Wang, Rui Wang, Yudong Wang, Ziyi
Wang, Xingjian Wei, Qizhen Weng, Fan Wu, Yingtong Xiong, Chao Xu, Ruiliang Xu, Hang
Yan, Yirong Yan, Xiaogui Yang, Haochen Ye, Huaiyuan Ying, Jia Yu, Jing Yu, Yuhang Zang,
Chuyu Zhang, Li Zhang, Pan Zhang, Peng Zhang, Ruijie Zhang, Shuo Zhang, Songyang
Zhang, Wenjian Zhang, Wenwei Zhang, Xingcheng Zhang, Xinyue Zhang, Hui Zhao,
Qian Zhao, Xiaomeng Zhao, Fengzhe Zhou, Zaida Zhou, Jingming Zhuo, Yicheng Zou,
Xipeng Qiu, Yu Qiao, Dahua Lin

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Qwen Team, Alibaba Group*

f a Small Language Model

Gabriel Martín Blázquez* Guilherme Penedo

Agustín Piqueres Lajarín Vaibhav Srivastav

Clémentine Fourrier Ben Burtenshaw

Cyril Zakka Mathieu Morlon

2024-06-27

d7db9

age Models

at a Practical Size

Gemma Team, Google DeepMind[1]

Phi-3 Tech

e Language

Microsoft

August 5, 2025 Release Product

Introducing gpt-oss

gpt-oss-120b and gpt-oss-20b push the frontier of open-weight reasoning model

Explore on Hugging Face ↗ Read model card ↗

MiniMax-M1: Scaling Test-Time Efficiently with Lightning Attent

MiniMax[1]

Llama 4:
Leading Multimodal In

Llama 4 Behemoth

Llama 4 Maverick

Llama 4 Scout

deepseek

DeepSeek-V3.2: Pushing the Frontier of Open Large Language Models

DeepSeek-AI

research@deepseek.com

2025-05-15

efficien
hybrid models for
enterprise

Published 02 October 2025

modal,

Liquid

Try LFM • Docs • LEAP • Disco

MiniMax M2 & Agent: Ingenious in Simplicity

Access API → Coding Plan → Try Agent Now

NVIDIA

NVIDIA Nemotron 3: Efficient and

models—Nano, Super, and Ultra. These models deliver strong
apabilities. The Nemotron 3 family uses a Mixture-of-Experts
re to provide best-in-class throughput and context lengths of
dels are trained with NVFP4 and incorporate LatentMoE, a
lity. The two larger models also include MTP layers for faster

INTERN-S1: A SCIENTIFIC MULTIMODAL FOUNDATION MODEL

Intern-S1 Team, Shanghai AI Laboratory

foundation models—together. We're
ratch, and sharing and programmatically
e, the data, the experiments, the
ite anyone who shares our vision of open
nd contribute, whether you want to try out
ithm, dataset, evaluation…there is a lot to

ERNIE 4.5 Technical Repo

KIMI K2: OPEN AGENTIC INTELLIGENCE

TECHNICAL REPORT OF KIMI K2

Kimi Team

oundation models
lixture-of-Experts
taining 424B total
MoE architecture,

of Maryland
here.

2025-12-22 · Research

GLM-4.7: Advancing the Coding Capability

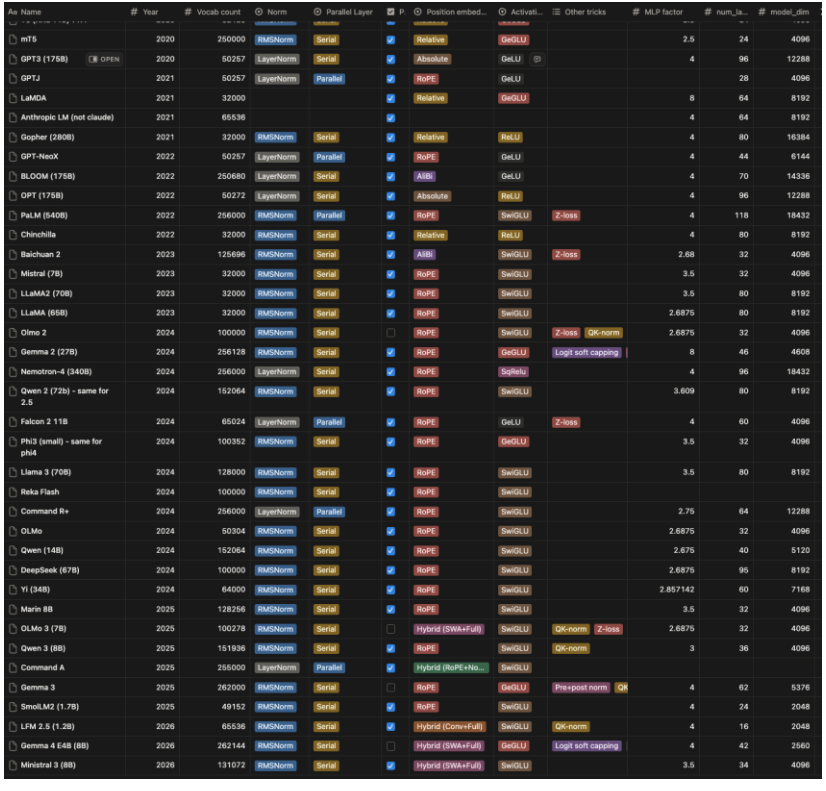
Try it at Z.ai ↗ Call it at Z.ai ↗ Z.ai Coding Plan ↗ GitHub ↗ HuggingFace ↗ Tech Report ↗

Step-3 is Large yet Affordable: Model-system Co-design for Cost-effective Decoding

StepFun Inc.

Let’s look at the data (on dense architectures)

Learn from the many other models (and papers) out there



Name	Year	Vocab count	Norm	Parallel Layer	P	Position embed...	Activati...	Other tricks	MLP factor	num_la...	model_dim
mT5	2020	250000	RMSNorm	Serial	☑	Relative	GeGLU		2.5	24	4096
GPT3 (175B) OPEN	2020	50257	LayerNorm	Serial	☑	Absolute	GeLU		4	96	12288
GPTJ	2021	50257	LayerNorm	Parallel	☑	RoPE	GeLU			28	4096
LaMDA	2021	32000			☑	Relative	GeGLU		8	64	8192
Anthropic LM (not claude)	2021	65536			☑				4	64	8192
Gopher (280B)	2021	32000	RMSNorm	Serial	☑	Relative	ReLU		4	80	16384
GPT-NeoX	2022	50257	LayerNorm	Parallel	☑	RoPE	GeLU		4	44	6144
BLOOM (175B)	2022	250680	LayerNorm	Serial	☑	AliBi	GeLU		4	70	14336
OPT (175B)	2022	50272	LayerNorm	Serial	☑	Absolute	ReLU		4	96	12288
PaLM (540B)	2022	256000	RMSNorm	Parallel	☑	RoPE	SwiGLU	Z-loss	4	118	18432
Chinchilla	2022	32000	RMSNorm	Serial	☑	Relative	ReLU		4	80	8192
Baichuan 2	2023	125696	RMSNorm	Serial	☑	AliBi	SwiGLU	Z-loss	2.68	32	4096
Mistral (7B)	2023	32000	RMSNorm	Serial	☑	RoPE	SwiGLU		3.5	32	4096
LLaMA2 (70B)	2023	32000	RMSNorm	Serial	☑	RoPE	SwiGLU		3.5	80	8192
LLaMA (65B)	2023	32000	RMSNorm	Serial	☑	RoPE	SwiGLU		2.6875	80	8192
Olmo 2	2024	100000	RMSNorm	Serial	☐	RoPE	SwiGLU	Z-loss QK-norm	2.6875	32	4096
Gemma 2 (27B)	2024	256128	RMSNorm	Serial	☑	RoPE	GeGLU	Logit soft capping	8	46	4608
Nemotron-4 (340B)	2024	256000	LayerNorm	Serial	☑	RoPE	SqReLu		4	96	18432
Qwen 2 (72b) - same for 2.5	2024	152064	RMSNorm	Serial	☑	RoPE	SwiGLU		3.609	80	8192
Falcon 2 11B	2024	65024	LayerNorm	Parallel	☑	RoPE	GeLU	Z-loss	4	60	4096
Phi3 (small) - same for phi4	2024	100352	RMSNorm	Serial	☑	RoPE	GeGLU		3.5	32	4096
Llama 3 (70B)	2024	128000	RMSNorm	Serial	☑	RoPE	SwiGLU		3.5	80	8192
Reka Flash	2024	100000	RMSNorm	Serial	☑	RoPE	SwiGLU				
Command R+	2024	256000	LayerNorm	Parallel	☑	RoPE	SwiGLU		2.75	64	12288
OLMo	2024	50304	RMSNorm	Serial	☑	RoPE	SwiGLU		2.6875	32	4096
Qwen (14B)	2024	152064	RMSNorm	Serial	☑	RoPE	SwiGLU		2.675	40	5120
DeepSeek (67B)	2024	100000	RMSNorm	Serial	☑	RoPE	SwiGLU		2.6875	95	8192
Yi (34B)	2024	64000	RMSNorm	Serial	☑	RoPE	SwiGLU		2.857142	60	7168
Marin 8B	2025	128256	RMSNorm	Serial	☑	RoPE	SwiGLU		3.5	32	4096
OLMo 3 (7B)	2025	100278	RMSNorm	Serial	☐	Hybrid (SWA+Full)	SwiGLU	QK-norm Z-loss	2.6875	32	4096
Qwen 3 (8B)	2025	151936	RMSNorm	Serial	☑	RoPE	SwiGLU	QK-norm	3	36	4096
Command A	2025	255000	LayerNorm	Parallel	☑	Hybrid (RoPE+No...	SwiGLU				
Gemma 3	2025	262000	RMSNorm	Serial	☐	RoPE	GeGLU	Pre+post norm QK	4	62	5376
SmolLM2 (1.7B)	2025	49152	RMSNorm	Serial	☑	RoPE	SwiGLU		4	24	2048
LFM 2.5 (1.2B)	2026	65536	RMSNorm	Serial	☑	Hybrid (Conv+Full)	SwiGLU	QK-norm	4	16	2048
Gemma 4 E4B (8B)	2026	262144	RMSNorm	Serial	☐	Hybrid (SWA+Full)	GeGLU	Logit soft capping	4	42	2560
Ministral 3 (8B)	2026	131072	RMSNorm	Serial	☑	Hybrid (SWA+Full)	SwiGLU		3.5	34	4096

We will talk through many major architecture and hyperparameter variants.

- What do all these models have in common?
- What parts vary?
- What can we learn from this?

What are we going to cover?

Common architecture variations

- Activations, FFN
- Attention variants
- Position embeddings

Standard blocks

Hyperparameters that (do or don't) matter

- What is ff_dim? Do multi_head dims always sum to model_dim?
- How many vocab elements?

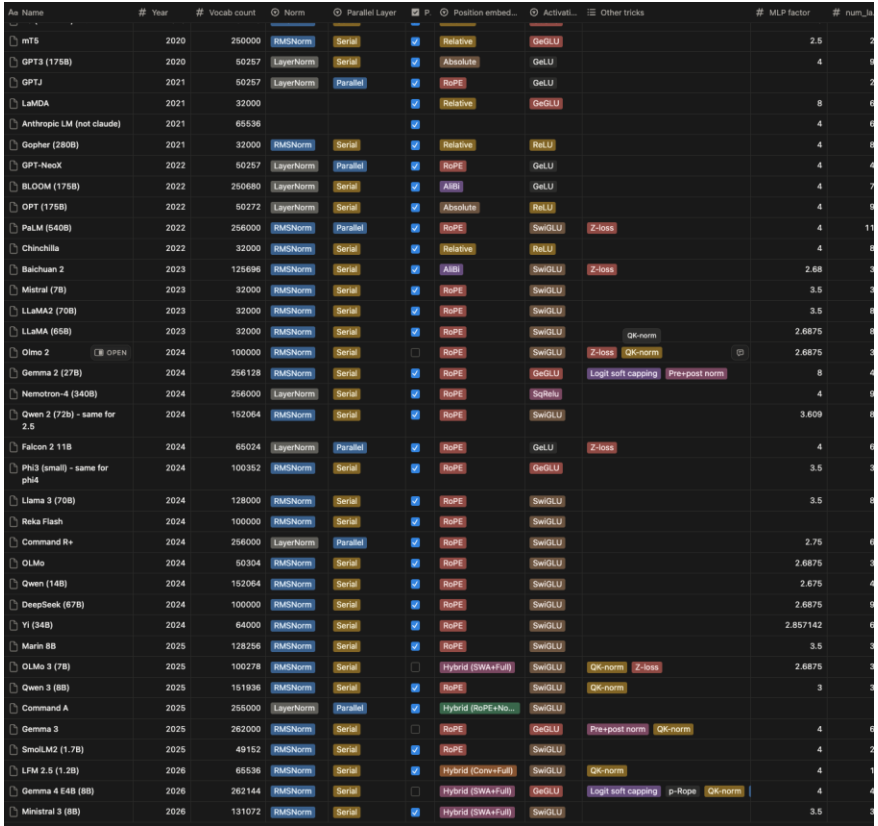
Stability tricks

How models can train stably (this have a connection with the architectures we can chose

Requirement of architectures:
1. Must learn from data (generalize
2. Must be efficient of the GPUs
3. Has not to blow up (when training)

Architecture variations..

Let’s think about the core architecture piece



Name	Year	Vocab count	Norm	Parallel Layer	P.	Position embed...	Activati...	Other tricks	MLP factor	num_la
mT5	2020	250000	RMSNorm	Serial	✓	Relative	GeGLU		2.5	
GPT3 (175B)	2020	50257	LayerNorm	Serial	✓	Absolute	GeLU		4	
GPTJ	2021	50257	LayerNorm	Parallel	✓	RoPE	GeLU			
LaMDA	2021	32000			✓	Relative	GeGLU		8	
Anthropic LM (not claude)	2021	65536			✓				4	
Gopher (280B)	2021	32000	RMSNorm	Serial	✓	Relative	ReLU		4	
GPT-NeoX	2022	50257	LayerNorm	Parallel	✓	RoPE	GeLU		4	
BLOOM (175B)	2022	250680	LayerNorm	Serial	✓	ALiBi	GeLU		4	
OPT (175B)	2022	50272	LayerNorm	Serial	✓	Absolute	ReLU		4	
PaLM (540B)	2022	256000	RMSNorm	Parallel	✓	RoPE	SwiGLU	Z-loss	4	
Chinchilla	2022	32000	RMSNorm	Serial	✓	Relative	ReLU		4	
Baichuan 2	2023	125696	RMSNorm	Serial	✓	ALiBi	SwiGLU	Z-loss	2.68	
Mistral (7B)	2023	32000	RMSNorm	Serial	✓	RoPE	SwiGLU		3.5	
LLaMA2 (70B)	2023	32000	RMSNorm	Serial	✓	RoPE	SwiGLU		3.5	
LLaMA (65B)	2023	32000	RMSNorm	Serial	✓	RoPE	SwiGLU	QK-norm	2.6875	
Olmo 2 OPEN	2024	100000	RMSNorm	Serial		RoPE	SwiGLU	Z-loss QK-norm	2.6875	
Gemma 2 (27B)	2024	256128	RMSNorm	Serial	✓	RoPE	GeGLU	Logit soft capping Pre+post norm	8	
Nemotron-4 (340B)	2024	256000	LayerNorm	Serial	✓	RoPE	SqRelu		4	
Qwen 2 (72b) - same for 2.5	2024	152064	RMSNorm	Serial	✓	RoPE	SwiGLU		3.609	
Falcon 2 11B	2024	65024	LayerNorm	Parallel	✓	RoPE	GeLU	Z-loss	4	
Phi3 (small) - same for phi4	2024	100352	RMSNorm	Serial	✓	RoPE	GeGLU		3.5	
Llama 3 (70B)	2024	128000	RMSNorm	Serial	✓	RoPE	SwiGLU		3.5	
Reka Flash	2024	100000	RMSNorm	Serial	✓	RoPE	SwiGLU			
Command R+	2024	256000	LayerNorm	Parallel	✓	RoPE	SwiGLU		2.75	
OLMo	2024	50304	RMSNorm	Serial	✓	RoPE	SwiGLU		2.6875	
Qwen (14B)	2024	152064	RMSNorm	Serial	✓	RoPE	SwiGLU		2.675	
DeepSeek (67B)	2024	100000	RMSNorm	Serial	✓	RoPE	SwiGLU		2.6875	
Yi (34B)	2024	64000	RMSNorm	Serial	✓	RoPE	SwiGLU		2.857142	
Marin 8B	2025	128256	RMSNorm	Serial	✓	RoPE	SwiGLU		3.5	
OLMo 3 (7B)	2025	100278	RMSNorm	Serial		Hybrid (SWA+Full)	SwiGLU	QK-norm Z-loss	2.6875	
Qwen 3 (8B)	2025	151936	RMSNorm	Serial	✓	RoPE	SwiGLU	QK-norm	3	
Command A	2025	255000	LayerNorm	Parallel	✓	Hybrid (RoPE+No...	SwiGLU			
Gemma 3	2025	262000	RMSNorm	Serial		RoPE	GeGLU	Pre+post norm QK-norm	4	
SmolLM2 (1.7B)	2025	49152	RMSNorm	Serial	✓	RoPE	SwiGLU		4	
LFM 2.5 (1.2B)	2026	65536	RMSNorm	Serial	✓	Hybrid (Conv+Full)	SwiGLU	QK-norm	4	
Gemma 4 E4B (8B)	2026	262144	RMSNorm	Serial		Hybrid (SWA+Full)	GeGLU	Logit soft capping p-Rope QK-norm	4	
Ministral 3 (8B)	2026	131072	RMSNorm	Serial	✓	Hybrid (SWA+Full)	SwiGLU		3.5	

After LLama 2 everything look like a variation
Now in 2025 we see major variations.

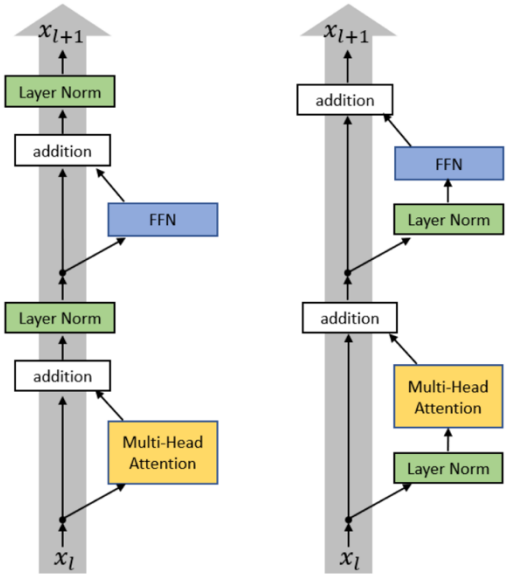
High level view:

- Dominance of ‘LLaMA-like’ architectures
- Trends over the years (QK-norm, Hybrid attention)

Probably the only thing Transformer Authors did wrong is where to put the
Layer Norm.
Layer Norm: Operazione in cui viene normalizzato ogni token separatamente, e viene anche imarata una traslazione comune a tutti i token.

Pre-vs-post norm

The one thing *everyone* agrees on (in 2024)



Post-LN Transformer	Pre-LN Transformer
$x_{l,i}^{post,1} = \text{MultiHeadAtt}(x_{l,i}^{post}, [x_{l,1}^{post}, \cdots, x_{l,n}^{post}])$	$x_{l,i}^{pre,1} = \text{LayerNorm}(x_{l,i}^{pre})$
$x_{l,i}^{post,2} = x_{l,i}^{post} + x_{l,i}^{post,1}$	$x_{l,i}^{pre,2} = \text{MultiHeadAtt}(x_{l,i}^{pre,1}, [x_{l,1}^{pre,1}, \cdots, x_{l,n}^{pre,1}])$
$x_{l,i}^{post,3} = \text{LayerNorm}(x_{l,i}^{post,2})$	$x_{l,i}^{pre,3} = x_{l,i}^{pre} + x_{l,i}^{pre,2}$
$x_{l,i}^{post,4} = \text{ReLU}(x_{l,i}^{post,3} W^{1,l} + b^{1,l}) W^{2,l} + b^{2,l}$	$x_{l,i}^{pre,4} = \text{LayerNorm}(x_{l,i}^{pre,3})$
$x_{l,i}^{post,5} = x_{l,i}^{post,3} + x_{l,i}^{post,4}$	$x_{l,i}^{pre,5} = \text{ReLU}(x_{l,i}^{pre,4} W^{1,l} + b^{1,l}) W^{2,l} + b^{2,l}$
$x_{l+1,i}^{post} = \text{LayerNorm}(x_{l,i}^{post,5})$	$x_{l+1,i}^{pre} = x_{l,i}^{pre,5} + x_{l,i}^{pre,3}$
	Final LayerNorm: $x_{Final,i}^{pre} \leftarrow \text{LayerNorm}(x_{L+1,i}^{pre})$

Set up LayerNorm so that it doesn't affect the main residual signal path (on the left)

Transformer originale
Post-Norm o Residual Norm
(ovvero Norm fatta su
Y=X+Att(X)

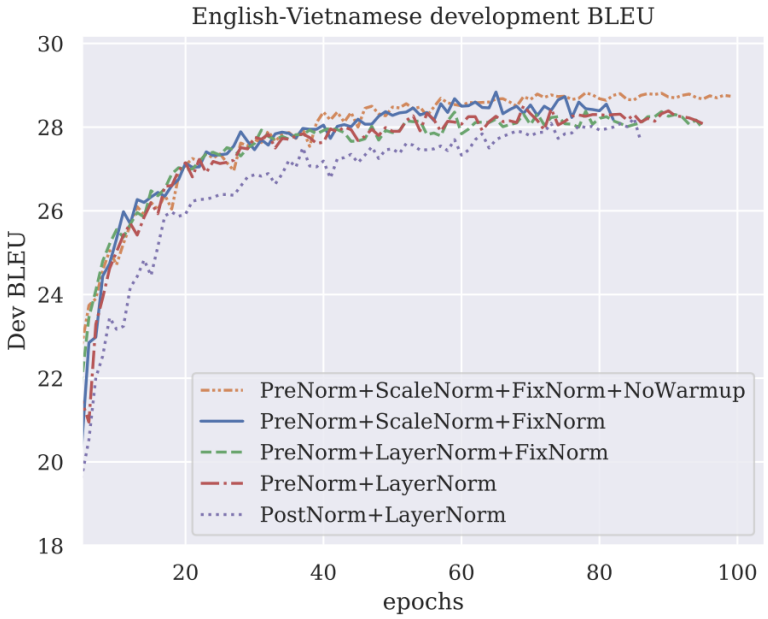
Figure from Xiong 2020

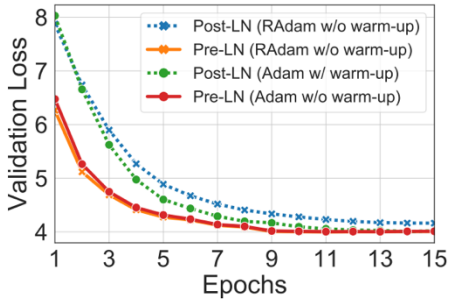
Alternativa Pre-Norm fuori dal residual stream e prima del FFN

Almost all modern LMs use pre-norm (but BERT was post-norm)

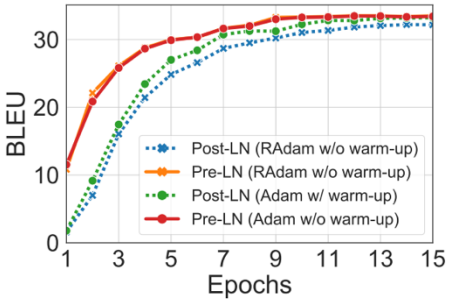
(One somewhat funny exception – OPT350M. I don't know why this is post-norm)

Pre-vs-post-norm, the data

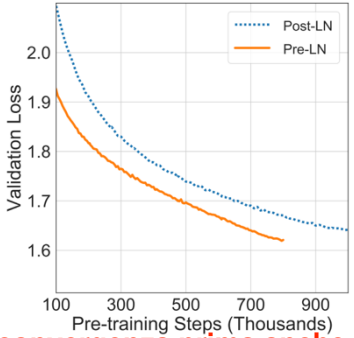




(a) Validation Loss (IWSLT)



(b) BLEU (IWSLT)



(a) Validation Loss on BERT

In pratica Post-Norm inizialmente è stato introdotto perche raggiunge convergenza prima anche senza warmup iniziale dei transformer ovvero un learning rate inizialmente piccolo vista l'instabilita iniziale

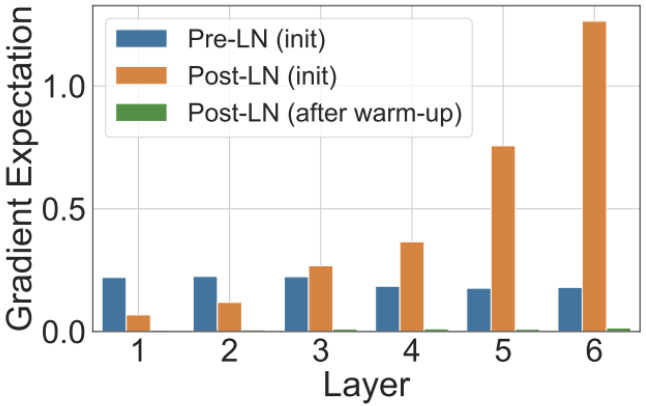
Salazar and Ngyuen 2019

Figure from Xiong 2020

Ma c'è anche delle altre consequenze che vengono fuori piu la rete diventa profonda
Per esempio che "Keep your Residual stream clean" fa si che il gradiente non esploda piu la rete diventa profonda, ma in generale ci sono anche altri motivi di stabilita

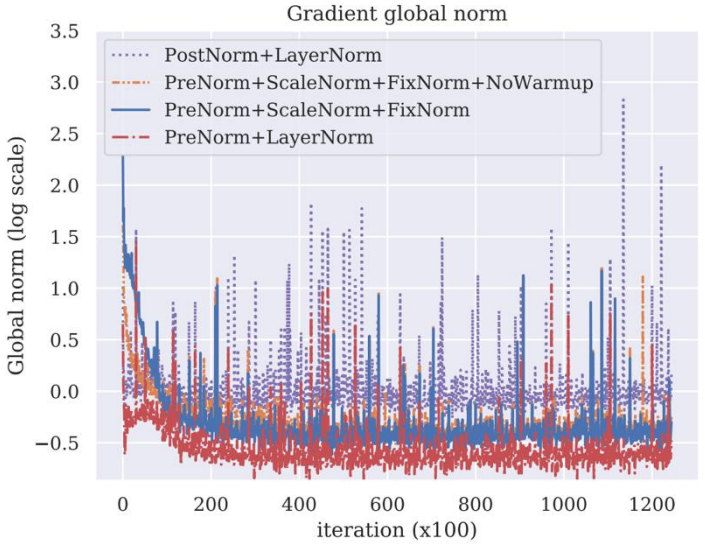
Pre-vs-post norm, explanations?

Gradient attenuation [Xiong 2020]



(a) W^1 in the FFN sub-layers

Gradient spikes [Salazar and Ngyuen]



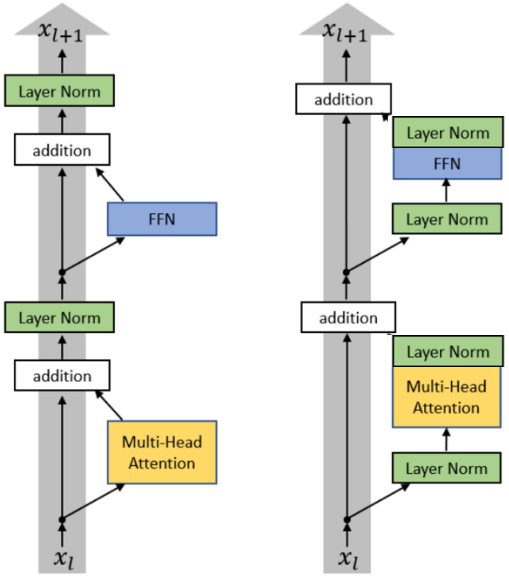
Original stated advantage– removing warmup.
Today – stability and larger LRs for large networks

New things – ‘double’ norm or non-residual postnorm

If putting LayerNorms in residual streams is bad.. Why not post-norm outside the stream?

Adesso altri modelli tengono comunque la norm fuori dallo stream ma li mettono un po ovunque nel non-residual per migliorare la stabilita

Post - Norm se solo dopo l’attention, DOuble se prima o dopo



Recent models: Grok, Gemma 2. Olmo 2 *only* does non-residual post norm

LayerNorm vs RMSNorm

Original transformer: **LayerNorm** – normalizes the mean and variance across d_{model}

$$y = \frac{x - \mathrm{E}[x]}{\sqrt{\mathrm{Var}[x] + \epsilon}} * \gamma + \beta$$

Notable models:

GPT3/2/1, OPT, GPT-J, BLOOM

All modern models fanno solo scaling down senza centrare per la media e senza imparare un termine di bias

Many modern LMs: **RMSNorm** – does not subtract mean or add a bias term

$$y = \frac{x}{\sqrt{||x||_2^2 + \varepsilon}} * \gamma$$

Notable models:

LLaMA-family, PaLM, Chinchilla, T5

Why RMSNorm?

The idea is, since we want to efficiently use our GPU and usually we are memory bounded, we can reduce operations if they doesn't help so much

Modern explanation – it's faster (and just as good).

- **Fewer operations** (no mean calculation)
- **Fewer parameters** (no bias term to store)

$$y = \frac{x - \mathrm{E}[x]}{\sqrt{\mathrm{Var}[x] + \epsilon}} * \gamma + \beta$$

Does this explanation make sense?

Operator class	% flop
△ Tensor contraction	99.80
□ Stat. normalization	0.17
○ Element-wise	0.03

Matrix multiplies are the *vast* majority of FLOPs (and memory)

Il fatto non è nemmeno che stiamo riducendo i FLOP, infatti queste due operazioni account solo per il 0,17% dei FLOP
IL punto è che riduciamo il runtime (queste operazione prendono fino al 25%!

[Ivanov et al 2023]

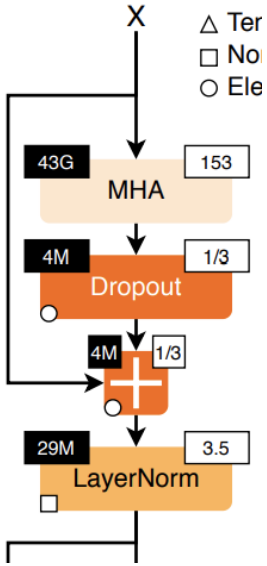
Why RMSNorm (2)

Important lesson: FLOPS are not runtime! (we will discuss this in far more detail later)

Il punto è che muoveremmo un sacco di volte avanti indietro i dati dalla memoria lenta alla veloce

Operator class	% flop	% Runtime
△ Tensor contraction	99.80	61.0
□ Stat. normalization	0.17	25.5
○ Element-wise	0.03	13.5

RMSNorm can still matter due to the importance of *data movement*



Left top (”43G”) is FLOPS
Right top (“153”) is the FLOP-to-memory ratio

[Ivanov et al 2023]

RMSNorm - validation

RMSNorm runtime (and surprisingly, perf) gains have been seen in papers

Model	Params	Ops	Step/s	Early loss	Final loss	SGLUE	XSum	WebQ	WMT EnDe
Vanilla Transformer	$223M$	$11.1T$	3.50	2.182 ± 0.005	1.838	71.66	17.78	23.02	26.62
RMS Norm	$223M$	$11.1T$	3.68	2.167 ± 0.008	**1.821**	**75.45**	**17.94**	**24.07**	**27.14**
Rezero	$223M$	$11.1T$	3.51	2.262 ± 0.003	1.939	61.69	15.64	20.90	26.37
Rezero + LayerNorm	$223M$	$11.1T$	3.26	2.223 ± 0.006	1.858	70.42	17.58	23.02	26.29
Rezero + RMS Norm	$223M$	$11.1T$	3.34	2.221 ± 0.009	1.875	70.33	17.32	23.02	26.19
Fixup	$223M$	$11.1T$	2.95	2.382 ± 0.012	2.067	58.56	14.42	23.02	26.31

Narang et al 2020

More generally: dropping bias terms

Most modern transformers don't have bias terms.

Original Transformer:

$$\mathrm{FFN}(x) = \max(0, xW_1 + b_1)W_2 + b_2$$

Most implementations (if they're not gated):

$$FFN(x) = \sigma(xW_1)W_2$$

Stessa cosa per i biasis, visto che non aiutano poi cosi tanto

Reasons: memory (similar to RMSnorm) and optimization stability

LayerNorm: recap

- Basically everyone does non-residual norm (often prenorm)
  - Intuition – keep the good parts of residual connections
  - Observations – nicer gradient propagation, fewer spike
  - Some people add a second norm outside the residual stream

- Most people do RMSnorm
  - In practice, works as well as LayerNorm
  - But, has fewer parameters to move around, which saves on wallclock time
  - People more generally drop bias terms since the compute/param tradeoffs are not great.

Activations

A whole zoo of activations ..

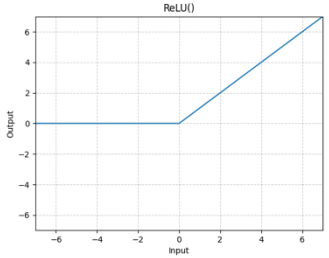
ReLU, GeLU, Swish, ELU, GLU, GeGLU, ReGLU, SeLU, SwiGLU, LiGLU

What are these things? What do people use? Does it matter?

A few of the common activations

ReLU

$$FF(x) = \max(0, xW_1)\, W_2$$



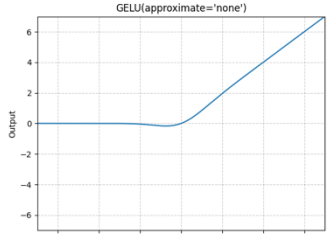
Notable models:

Original transformer, T5, Gopher, Chinchilla, OPT

Chinchilla il meglio di questo gruppo

GeLU

$$FF(x) = \mathrm{GELU}(xW_1)W_2$$
$$GELU(x) \coloneqq x\Phi(x)$$



Notable models:

GPT1/2/3, GPTJ, GPT-Neox, BLOOM

GELU is a smoother alternative to ReLU that allows small negative outputs, rather than setting every negative input to zero where Phi is the cumulative distribution function of a standard Gaussian

SwiGLU / GeGLU (next slide..)

Notable models:

Llama, PaLM,T5 v1.1, *most* models post 2023

Gated activations (*GLU)

GLUs modify the 'first part' of a FF layer

standard ReLU

$$FF(x) = \max(0, xW_1)\, W_2$$

Instead of a linear + ReLU, augment the above with an (entrywise) linear term

Qua moltiplico per un gate (V) same size

$$\max(0, xW_1) \rightarrow \max(0, xW_1) \otimes (xV)$$

This gives the gated variant (ReGLU) – note that we have an extra parameter (V)

$$\mathrm{FF}_{\mathrm{ReGLU}}(x) = (\max(0, xW_1) \otimes xV)\, W_2$$

Gated variants of standard FF layers

GeGLU GeLU with a Gate

$$\mathrm{FFN}_{\mathrm{GEGLU}}(x, W, V, W_2) = (\mathrm{GELU}(xW) \otimes xV)W_2$$

Notable models:

T5 v1.1, mT5, LaMDA, Phi3, Gemma 2, Gemma 3, Gemma 4

SwiGLU (swish is $x * \mathrm{sigmoid}(x)$)

Also here we add a gate

$$\mathrm{FFN}_{\mathrm{SwiGLU}}(x, W, V, W_2) = (\mathrm{Swish}_1(xW) \otimes xV)W_2$$

SwiGLU is probably the dominant right now

Notable models:

LLaMa 1/2/3, PaLM, Mistral, OlMo, *most models post 2023*

Note: Gated models use smaller dimensions for the d_{ff} by 2/3

Not an iron rule but people now compensate the fact that you add another matrix with smaller FF dimensions (2/3 of the original) to keep the same number of parameters. (If you do the maths, it comes out to 2/3)

Do gated linear units work?

Yes, fairly consistently so.

	Score Average	CoLA MCC	SST-2 Acc
FFN_{ReLU}	83.80	51.32	94.04
FFN_{GELU}	83.86	53.48	94.04
$\text{FFN}_{\text{Swish}}$	83.60	49.79	93.69
FFN_{GLU}	84.20	49.16	94.27
$\text{FFN}_{\text{GEGLU}}$	84.12	53.65	93.92
$\text{FFN}_{\text{Bilinear}}$	83.79	51.02	**94.38**
$\text{FFN}_{\text{SwiGLU}}$	84.36	51.59	93.92
$\text{FFN}_{\text{ReGLU}}$	**84.67**	**56.16**	**94.38**
[Raffel et al., 2019]	83.28	53.84	92.68
ibid. stddev.	0.235	1.111	0.569

Shazeer 2020

How you can see GLU works much better (at the same size of parameters)

Do gated linear units work (2)?

Yes, with other works corroborating Shazeer 2020

Model	Params	Ops	Step/s	Early loss	Final loss	SGLUE	XSum	WebQ
Vanilla Transformer	$223M$	$11.1T$	3.50	2.182 ± 0.005	1.838	71.66	17.78	23.02
GeLU	$223M$	$11.1T$	3.58	2.179 ± 0.003	1.838	**75.79**	**17.86**	**25.13**
Swish	$223M$	$11.1T$	3.62	2.186 ± 0.003	1.847	**73.77**	17.74	**24.34**
ELU	$223M$	$11.1T$	3.56	2.270 ± 0.007	1.932	67.83	16.73	23.02
GLU	$223M$	$11.1T$	3.59	2.174 ± 0.003	**1.814**	**74.20**	**17.42**	24.34
GeGLU	$223M$	$11.1T$	3.55	2.130 ± 0.006	**1.792**	**75.96**	**18.27**	**24.87**
ReGLU	$223M$	$11.1T$	3.57	2.145 ± 0.004	**1.803**	**76.17**	**18.36**	**24.87**
SeLU	$223M$	$11.1T$	3.55	2.315 ± 0.004	1.948	68.76	16.76	22.75
SwiGLU	$223M$	$11.1T$	3.53	2.127 ± 0.003	**1.789**	**76.00**	**18.20**	**24.34**
LiGLU	$223M$	$11.1T$	3.59	2.149 ± 0.005	**1.798**	**75.34**	**17.97**	**24.34**
Sigmoid	$223M$	$11.1T$	3.63	2.291 ± 0.019	1.867	**74.31**	17.51	23.02
Softplus	$223M$	$11.1T$	3.47	2.207 ± 0.011	1.850	**72.45**	17.65	**24.34**

Narang et al 2020

Gating, activations

- **Many variations (ReLU, GeLU, *GLU) across models.**

- ***GLU isn't *necessary* for a working model (see GPT3), but it's rare to see others..**

  Some outlier models..

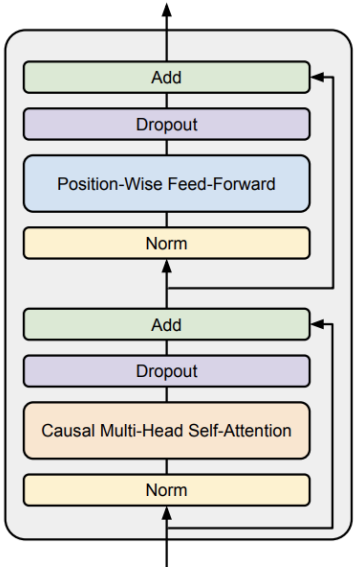
  Nemotron 340B (Squared ReLU)

- **Evidence points towards somewhat consistent gains from Swi/GeGLU**

Serial vs Parallel layers

Can we parallelize the MLP and the Attention to avoid to wait?

Normal transformer blocks are *serial* – they compute attention, then the MLP



Could we parallelize the transformer block?

Parallel layers

A few models (GPTJ, PaLM, GPT-NeoX) do parallel layers. Originally in GPT-J

Parallel Layers – We use a "parallel" formulation in each Transformer block (Wang & Komatsuzaki, 2021), rather than the standard "serialized" formulation. Specifically, the standard formulation can be written as:

$$y = x + \text{MLP}(\text{LayerNorm}(x + \text{Attention}(\text{LayerNorm}(x)))$$

Whereas the parallel formulation can be written as:

$$y = x + \text{MLP}(\text{LayerNorm}(x)) + \text{Attention}(\text{LayerNorm}(x))$$

The parallel formulation results in roughly 15% faster training speed at large scales, since the MLP and Attention input matrix multiplications can be fused. Ablation experiments showed a small quality degradation at 8B scale but no quality degradation at 62B scale, so we extrapolated that the effect of parallel layers should be quality neutral at the 540B scale.

If implemented right, LayerNorm can be shared, and matrix multiplies can be fused

Recent Models: Cohere Command A, Falcon 2 11B, Command R+

Answer: You could and there are advantages but the serialization now it's not a bottleneck and you are losing the depht

Summary: architectures

Pre-vs-post norm:

- Everyone does non-residual norm (except OPT350M), likely with good reason.

Layer vs RMSnorm:

- RMSnorm has clear compute wins, sometimes even performance

Gating:

- GLUs are consensus now

Serial vs parallel layers:

- Most models now use serial layers

Name	Year	Norm	Parallel Layer	Pre-norm	Position embedding	Activations
Original transformer	2017	LayerNorm	Serial	☐	Sine	ReLU
GPT	2018	LayerNorm	Serial	☐	Absolute	GeLU
T5 (11B)	2019	RMSNorm	Serial	☑	Relative	ReLU
GPT2	2019	LayerNorm	Serial	☑	Absolute	GeLU
T5 (XXL 11B) v1.1	2020	RMSNorm	Serial	☑	Relative	GeGLU
mT5	2020	RMSNorm	Serial	☑	Relative	GeGLU
GPT3 (175B)	2020	LayerNorm	Serial	☑	Absolute	GeLU
GPTJ	2021	LayerNorm	Parallel	☑	RoPE	GeLU
LaMDA	2021			☑	Relative	GeGLU
Anthropic LM (not claude)	2021			☑		
Gopher (280B)	2021	RMSNorm	Serial	☑	Relative	ReLU
GPT-NeoX	2022	LayerNorm	Parallel	☑	RoPE	GeLU
BLOOM (175B)	2022	LayerNorm	Serial	☑	AliBi	GeLU
OPT (175B)	2022	LayerNorm	Serial	☑	Absolute	ReLU
PaLM (540B)	2022	RMSNorm	Parallel	☑	RoPE	SwiGLU
Chinchilla	2022	RMSNorm	Serial	☑	Relative	ReLU
Mistral (7B)	2023	RMSNorm	Serial	☑	RoPE	SwiGLU
LLaMA2 (70B)	2023	RMSNorm	Serial	☑	RoPE	SwiGLU
LLaMA (65B)	2023	RMSNorm	Serial	☑	RoPE	SwiGLU
GPT4	2023			☐		
Baichuan 2	2023	RMSNorm	Serial	☑	AliBi	SwiGLU
Olmo 2	2024	RMSNorm	Serial	☐	RoPE	SwiGLU
Gemma 2 (27B)	2024	RMSNorm	Serial	☑	RoPE	GeGLU
Nemotron-4 (340B)	2024	LayerNorm	Serial	☑	RoPE	SqRelu
Qwen 2 (72b) - same for 2.5	2024	RMSNorm	Serial	☑	RoPE	SwiGLU
Falcon 2 11B	2024	LayerNorm	Parallel	☑	RoPE	GeLU
Phi3 (small) - same for phi4	2024	RMSNorm	Serial	☑	RoPE	GeGLU
Llama 3 (70B)	2024	RMSNorm	Serial	☑	RoPE	SwiGLU
Reka Flash	2024	RMSNorm	Serial	☑	RoPE	SwiGLU
Command R+	2024	LayerNorm	Parallel	☑	RoPE	SwiGLU
OLMo	2024	RMSNorm	Serial	☑	RoPE	SwiGLU
Qwen (14B)	2024	RMSNorm	Serial	☑	RoPE	SwiGLU
DeepSeek (67B)	2024	RMSNorm	Serial	☑	RoPE	SwiGLU
Yi (34B)	2024	RMSNorm	Serial	☑	RoPE	SwiGLU
Mixtral of Experts	2024			☐		
Command A	2025	LayerNorm	Parallel	☑	Hybrid (RoPE+NoPE)	SwiGLU
Gemma 3	2025	RMSNorm	Serial	☐	RoPE	GeGLU
SmolLM2 (1.7B)	2025	RMSNorm	Serial	☑	RoPE	SwiGLU

Many variations in position embeddings

Sine embeddings: add sines and cosines that enable localization

$$Embed(x, i) = v_x + PE_{pos}$$

$$PE_{(pos,2i)} = sin(pos/10000^{2i/d_{\text{model}}})$$
$$PE_{(pos,2i+1)} = cos(pos/10000^{2i/d_{\text{model}}})$$

Notable models:

Original transformer

Absolute embeddings: add a position vector to the embedding

$$Embed(x, i) = v_x + u_i$$

Notable models:

GPT1/2/3, OPT

Relative embeddings: add a vector to the *attention computation*

$$e_{ij} = \frac{x_i W^Q (x_j W^K + a_{ij}^K)^T}{\sqrt{d_z}}$$

Notable models:

T5, Gopher, Chinchilla

Rope embeddings (next slides..)

Notable models:

GPTJ, PaLM, LLaMA

Most 2024+ models

RoPE: rotary position embeddings

High level thought process: a *relative* position embedding should be some $f(x, i)$ s.t.

$$\langle f(x,i), f(y,j) \rangle = g(x, y, i-j)$$

That is, the attention function *only* gets to depend on the relative position (i-j). How do existing embeddings not fulfill this goal?

- **Sine:** Has various cross-terms that are not relative

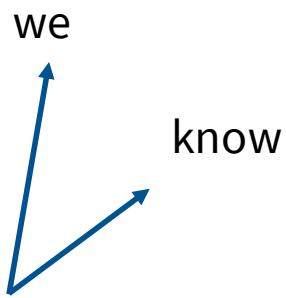
$$\langle Embed(x,i), Embed(y,i) \rangle = \langle v_x, v_y \rangle + \langle PE_i, v_y \rangle \ldots$$

- **Absolute:** obviously not relative
- **Relative embeddings:** $e_{ij} = \frac{x_i W^Q (x_j W^K + a_{ij}^K)^T}{\sqrt{d_z}}$ is not an inner product

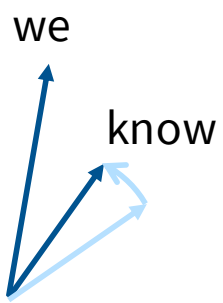
RoPE: rotary position embeddings

How can we solve this problem?

- We want our embeddings to be invariant to absolute position
- We know that inner products are invariant to arbitrary rotation.



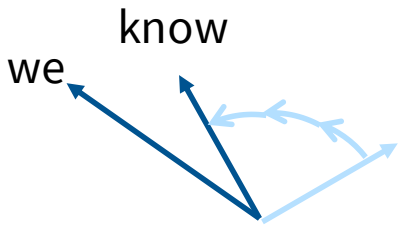
Position independent embedding



Embedding
“we know that”

Rotate we by ‘0 positions’

know by ‘1 positions’



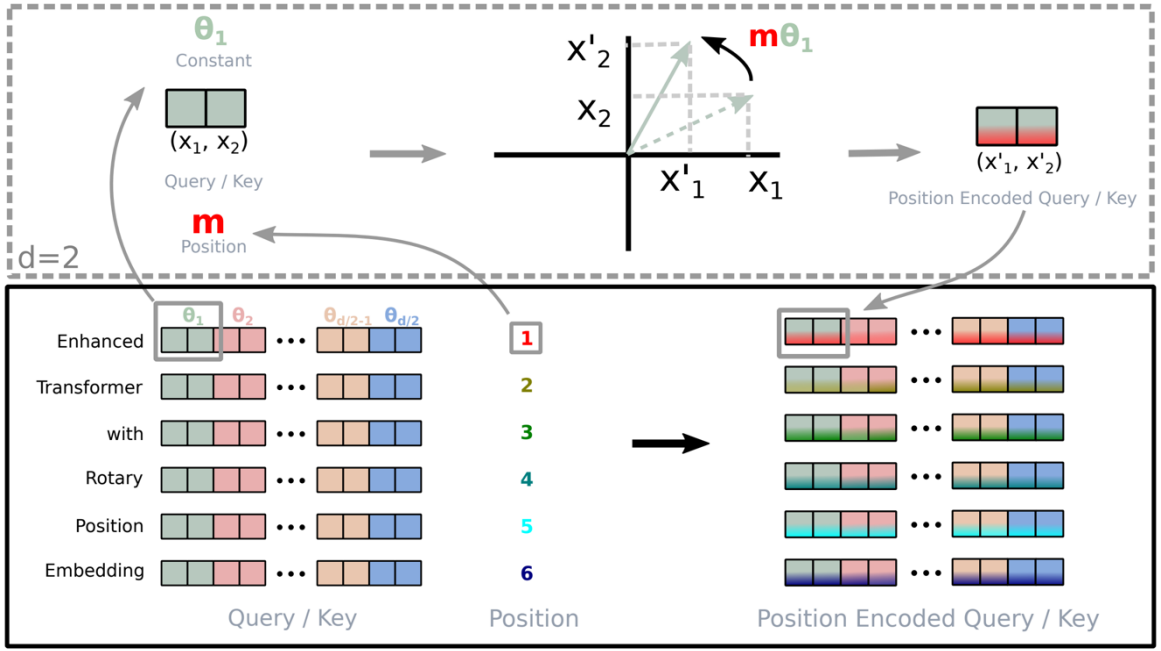
Embedding
“of course we know”

Rotate we by ‘2 positions’

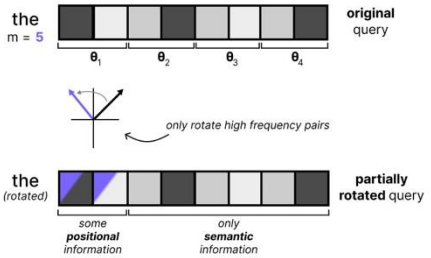
Rotate know by ‘3 positions’

RoPE: rotary position embeddings

There are many rotations, which one do you pick?



[Su et al 2021]



Gemma 4 alternative: just first 2

Just pair up the coordinates and rotate them in 2d (motivation: complex numbers)

The actual RoPE math

Multiply with sines and cosines

$$f_{\{q,k\}}(\boldsymbol{x}_m, m) = \boldsymbol{R}^d_{\Theta,m}\boldsymbol{W}_{\{q,k\}}\boldsymbol{x}_m \tag{14}$$

$$\boldsymbol{R}^d_{\Theta,m} = \begin{pmatrix} \cos m\theta_1 & -\sin m\theta_1 & 0 & 0 & \cdots & 0 & 0 \\ \sin m\theta_1 & \cos m\theta_1 & 0 & 0 & \cdots & 0 & 0 \\ 0 & 0 & \cos m\theta_2 & -\sin m\theta_2 & \cdots & 0 & 0 \\ 0 & 0 & \sin m\theta_2 & \cos m\theta_2 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & \cdots & \cos m\theta_{d/2} & -\sin m\theta_{d/2} \\ 0 & 0 & 0 & 0 & \cdots & \sin m\theta_{d/2} & \cos m\theta_{d/2} \end{pmatrix} \tag{15}$$

Difference with sine embeddings – not additive, no cross terms

Implementation and code for RoPE

Usual attention stuff

Get the RoPE matrix cos/sin

Multiply query/key inputs

```
query_states = self.q_proj(hidden_states)
key_states = self.k_proj(hidden_states)
value_states = self.v_proj(hidden_states)

# Flash attention requires the input to have the shape
# batch_size x seq_length x head_dim x hidden_dim
# therefore we just need to keep the original shape
query_states = query_states.view(bsz, q_len, self.num_heads, self.head_dim).transpose(1, 2)
key_states = key_states.view(bsz, q_len, self.num_key_value_heads, self.head_dim).transpose(1, 2)
value_states = value_states.view(bsz, q_len, self.num_key_value_heads, self.head_dim).transpose(1, 2)

cos, sin = self.rotary_emb(value_states, position_ids)
query_states, key_states = apply_rotary_pos_emb(query_states, key_states, cos, sin)
```

…

Same stuff as the usual multi-head self attention below

Note: embedding at *each attention operation* to enforce position invariance

Hyperparameters

Transformer hyperparameter questions you might have had in 224n..

- How much bigger should the feedforward size be compared to hidden size?
- How many heads, and should num_heads always divide hidden size?
- What should my vocab size be?

And other model setting questions

- Do people even regularize these huge LMs?
- How do people scale these models - very deep or very wide?

Surprising (?) consensus hyperparameter 1

Feedforward – model dimension ratio.

$$\mathrm{FFN}(x) = \max(0, xW_1 + b_1)W_2 + b_2$$

There are two dimensions that are relevant – the feedforward dim (d_{ff}) and model dim (d_{model}). What should their relationship be?

$$\boldsymbol{d_{ff} = 4\, d_{model}}$$

This is *almost always* true. There's just a few exceptions.

Exception #1 – GLU variants

Remember that GLU variants scale down by 2/3rd. This means most GLU variants have $d_{ff} = \frac{8}{3} d_{model}$. This is mostly what happens. Some notable such examples.

Model	d_{ff}/d_{model}
PaLM	4
Mistral 7B	3.5
LLaMA-2 70B	3.5
LLaMA 70B	2.68
Qwen 14B	2.67
DeepSeek 67B	2.68
Yi 34B	2.85
T5 v1.1	2.5

Models are roughly in this range, though PaLM, LLaMA2 and Mistral are slightly larger

Exception #2 – T5

As we have (and will) see, most LMs are have boring, conservative hyperparameters. One exception is T5 [Raffel et al 2020] which has some *very bold* settings.

In particular, for the 11B model, they set

$$d_{ff} = 65{,}536$$
$$d_{model} = 1024$$

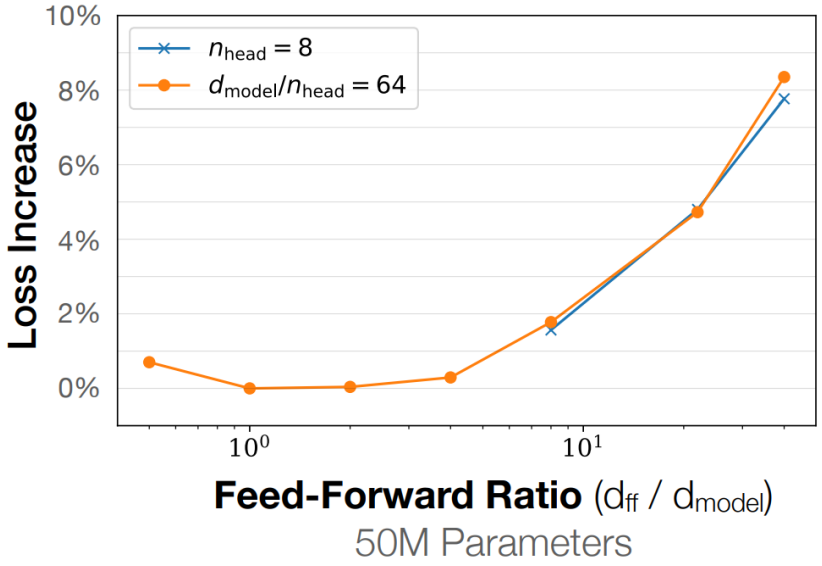
For an astounding 64-times multiplier.

> for "11B" we use $d_{\text{ff}} = 65{,}536$ with 128-headed attention producing a model with about 11 billion parameters. We chose to scale up d_{ff} specifically because modern accelerators (such as the TPUs we train our models on) are most efficient for large dense matrix multiplications like those in the Transformer's feed-forward networks.

Other, recent exceptions – Gemma 2 (8x), SmolLM/Gemma 3/Gemma 4 (4x, GLU)

Why this range of multipliers?

Empirically, there's a basin between 1-10 where this hyperparameter is near-optimal



[Kaplan+ 2020]

What can we learn from the model-dim hyperparam?

- The ‘default’ choices of $d_{ff} = 4d_{model}$ and $d_{ff} = 2.66d_{model}$ have worked well for nearly all modern LLMs.

- But T5 does show that even radical choices of $d_{ff} = 64d_{model}$ can work. This hyperparameter choice isn’t written in stone.

- That said, T5 has a follow-up model (T5 v1.1) that is ‘improved’ and uses a much more standard 2.5 multiplier on GeGLU, so the 64-times multiplier is likely suboptimal.

Surprising (?) consensus hyperparameter 2

Head-dim*num-heads to model-dim ratio. As a reminder, slide from 224n.

Multi-head self-attention is computationally efficient

- Even though we compute h many attention heads, it's not really more costly.
  - We compute $XQ \in \mathbb{R}^{n \times d}$, and then reshape to $\mathbb{R}^{n \times h \times d/h}$. (Likewise for XK, XV.)
  - Then we transpose to $\mathbb{R}^{h \times n \times d/h}$; now the head axis is like a batch axis.
  - Almost everything else is identical, and the **matrices are the same sizes.**

This doesn't *have to* be true: we can have head-dimensions > model-dim / num-heads.

But most models do follow this guideline

How many heads, whats the model dim?

Some examples of this hyperparameter

	Num heads	Head dim	Model dim	Ratio
GPT3	96	128	12288	1
T5	128	128	1024	16
T5 v1.1	64	64	4096	1
LaMDA	128	128	8192	2
PaLM	48	258	18432	1.48
LLaMA2	64	128	8192	1
Qwen 3.5 (27B)	24	256	5120	1.2

Most models have ratios around 1 – notable exceptions by some google models.

Aspect ratios

Should my model be deep or wide? *How* deep and how wide?

Most models are surprisingly consistent on this one too!

Sweet spot?

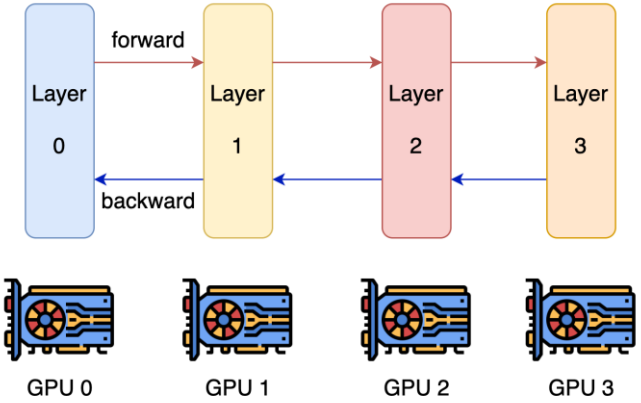
Model	d_{model}/n_{layer}
BLOOM	205
T5 v1.1	171
PaLM (540B)	156
GPT3/OPT/Mistral/Qwen /OLMo 3	128
LLaMA / LLaMA2	102
Gemma 3	87
Gemma 4	61
T5 (11B)	33

Considerations about aspect ratio

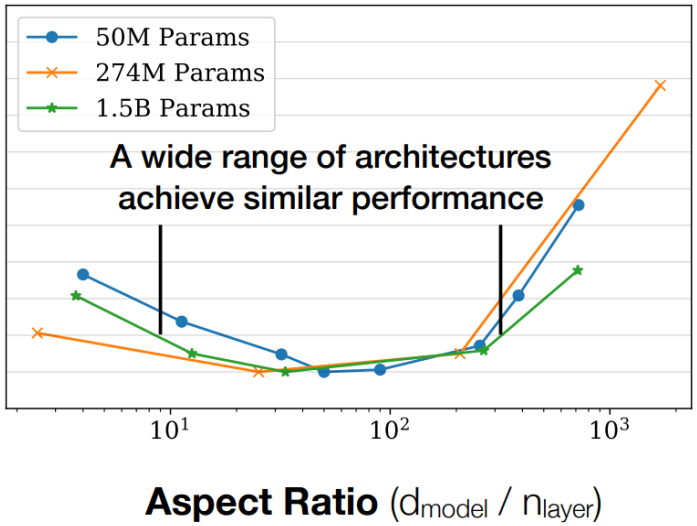
Extremely deep models are harder to parallelize and have higher latency

> **The Limits of Depth vs Width** We note an obvious limitation with our advice. Scaling depth has an obvious limiter, i.e., they are non-parallelizable across different machines or devices and every computation has to always wait for the previous layer. This is unlike width, which can be easily parallelizable over thousands or hundreds of thousands of devices. Within the limitation of scaling

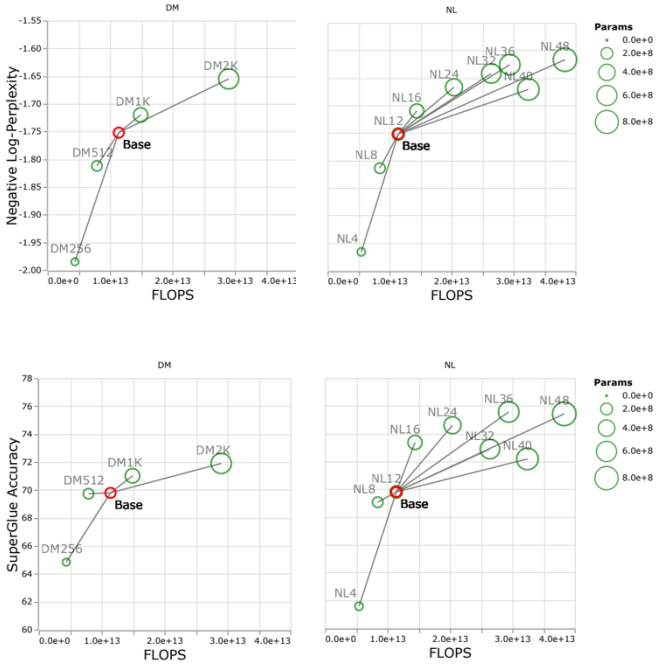
[Tay et al 2021]



Evidence on aspect ratio scaling



[Kaplan et al 2020]



[Tay et al 2021]

What are typical vocabulary sizes?

Monolingual models – 30-50k vocab

Model	Token count
Original transformer	37000
GPT	40257
GPT2/3	50257
T5/T5v1.1	32128
LLaMA	32000

Multilingual / production systems 100-250k

Model	Token count
mT5	250000
PaLM	256000
GPT4	100276
Gemma 4	262144
DeepSeek	100000
Qwen 15B	152064
Yi	64000

Monolingual vocabs don’t need to be huge, but multilingual ones do

Dropout and other regularization

Do we need regularization during pretraining?

Arguments against:

- There is *a lot* of data (trillions of tokens), more than parameters.
- SGD only does a single pass on a corpus (hard to memorize)

This is all quite reasonable.. but what do people do in practice?

Dropout and weight decay in practice

Model	Dropout*	Weight decay
Original transformer	0.1	0
GPT2	0.1	0.1
T5	0.1	0
GPT3	0.1	0.1
T5 v1.1	0	0
PaLM	0	(variable)
OPT	0.1	0.1
LLaMA	0	0.1
Qwen 14B	0.1	0.1

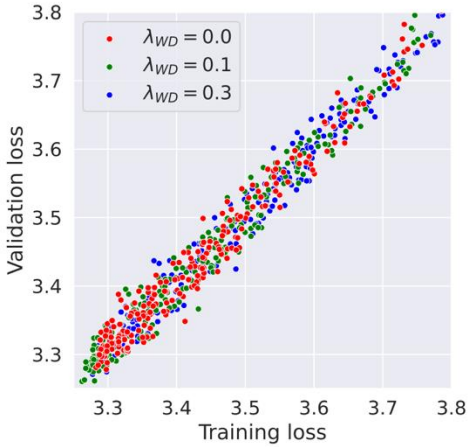
Many older models used dropout during pretraining

Newer models (except Qwen) rely only on weight decay

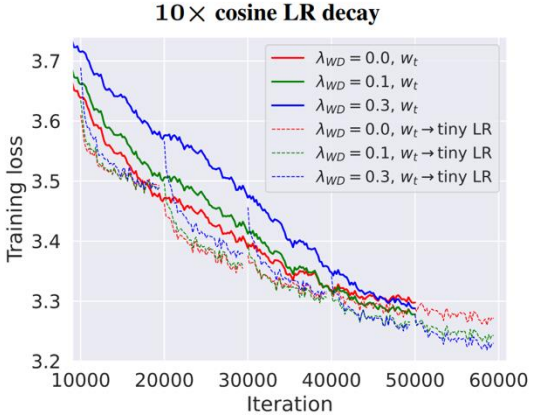
* Most of the times papers just don't discuss dropout. On open models, this closely matches not doing dropout. This may not be true of closed models.

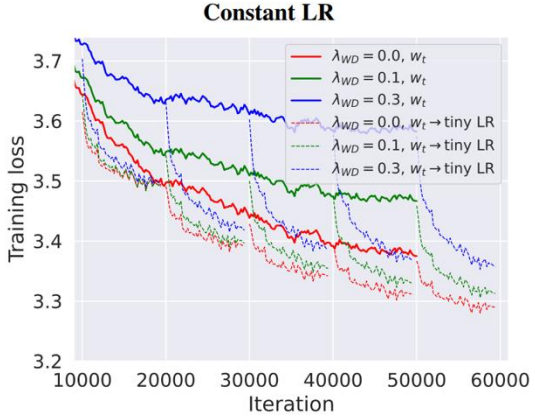
Why weight decay LLMs?

[Andriushchenko et al 2023] has interesting observations about LLM weight decay



It's not to control overfitting





Weight decay interacts with learning rates (cosine schedule)

Summary: hyperparameters

Feedforward

- Factor-of-4 rule of thumb (8/3 for GLUs) is standard (with some evidence)

Head dim

- Head dim*Num head = D model is standard – but low to no validation

Aspect ratio

- Wide range of ‘good’ values (100-200). Systems concerns dictate the value

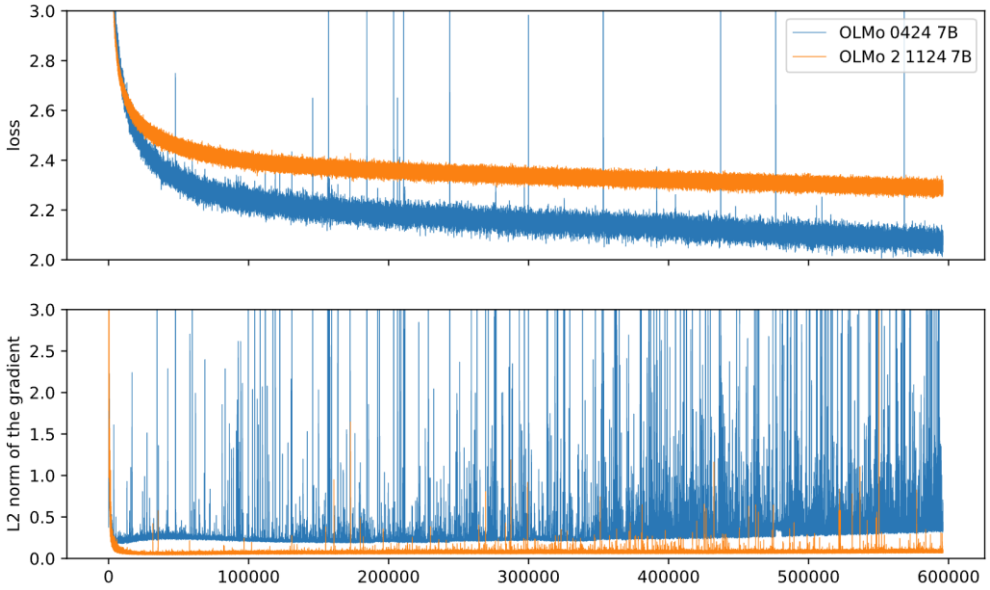
Regularization

- You still ‘regularize’ LMs but its effects are primarily on optimization dynamics

Name	Year	MLP factor	Aspect ratio (d/layer)	weight decay	drop_rate
Original transformer	2017	4	85	0	0.1
GPT	2018	4	64	0.1	0.1
T5 (11B)	2019	64	43	0	0.1
GPT2	2019	4	33	0.1	0.1
T5 (XXL 11B) v1.1	2020	2.5	171	0	0
mT5	2020	2.5	171	0	0
GPT3 (175B)	2020	4	128	0.1	0.1
GPTJ	2021		146	0.1	0
LaMDA	2021	8	128		
Anthropic LM (not claude)	2021	4	128		
Gopher (280B)	2021	4	205		
GPT-NeoX	2022	4	140	0.01	0
BLOOM (175B)	2022	4	205	0.1	0
OPT (175B)	2022	4	128	0.1	0.1
PaLM (540B)	2022	4	156		0
Chinchilla	2022	4	102		
Baichuan 2	2023	2.68	128	0.1	0
Mistral (7B)	2023	3.5	128	0.1	0
LLaMA2 (70B)	2023	3.5	102	0.1	0
LLaMA (65B)	2023	2.6875	102	0.1	0
GPT4	2023		0		
Olmo 2	2024	2.6875	128		
Gemma 2 (27B)	2024	8	100		
Nemotron-4 (340B)	2024	4	192		0
Qwen 2 (72b) - same for 2.5	2024	3.609	102		
Falcon 2 11B	2024	4	68	0.1	
Phi3 (small) - same for phi4	2024	3.5	128		
Llama 3 (70B)	2024	3.5	102		0
Reka Flash	2024		0		
Command R+	2024	2.75	192		
OLMo	2024	2.6875	128	0.1	0
Qwen (14B)	2024	2.675	128	0.1	0.1
DeepSeek (67B)	2024	2.6875	86	0.1	0
Yi (34B)	2024	2.857142	119	0.1	0
Mixtral of Experts	2024		0		
Command A	2025		0		
Gemma 3	2025	4	87		
SmolLM2 (1.7B)	2025	4	85		

Stability tricks

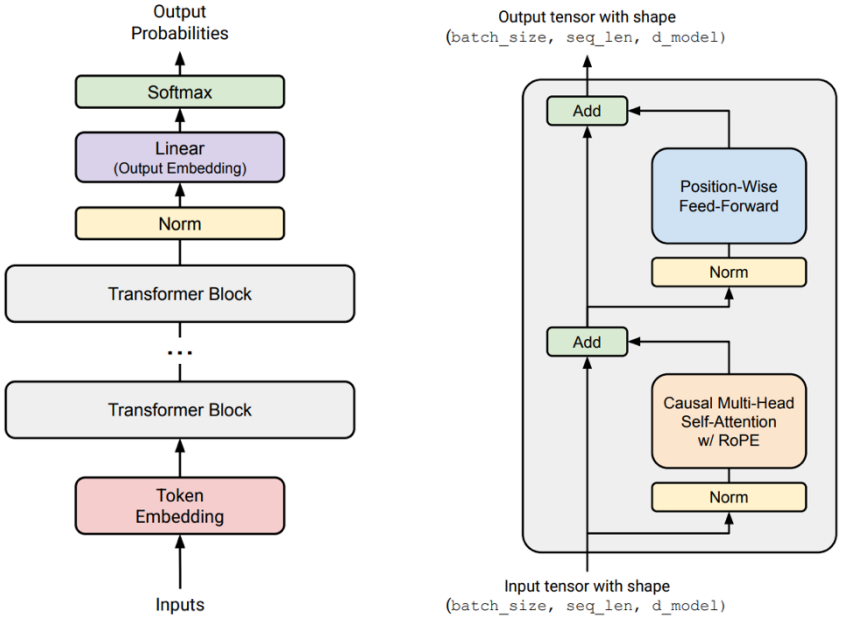
Recently, lots of attention on *stable training*



Don’t train models that look like the blue curve!

Where do the issues arise? Beware of softmaxes!

Softmaxes – can be ill-behaved due to exponentials / divison by zero



Output softmax stability – the ‘z-loss’

Recall the softmax calculation

$$\begin{aligned} \log(P(x)) &= \log\left(\frac{e^{U_r(x)}}{Z(x)}\right) \\ &= U_r(x) - \log(Z(x)) \\ Z(x) &= \Sigma_{r'=1}^{|V|} e^{U_{r'}(x)} \end{aligned}$$

$$\begin{aligned} L &= \sum_i \left[\log(P(x_i)) - \alpha(\log(Z(x_i)) - 0)^2\right] \\ &= \sum_i \left[\log(P(x_i)) - \alpha \log^2(Z(x_i))\right] \end{aligned}$$

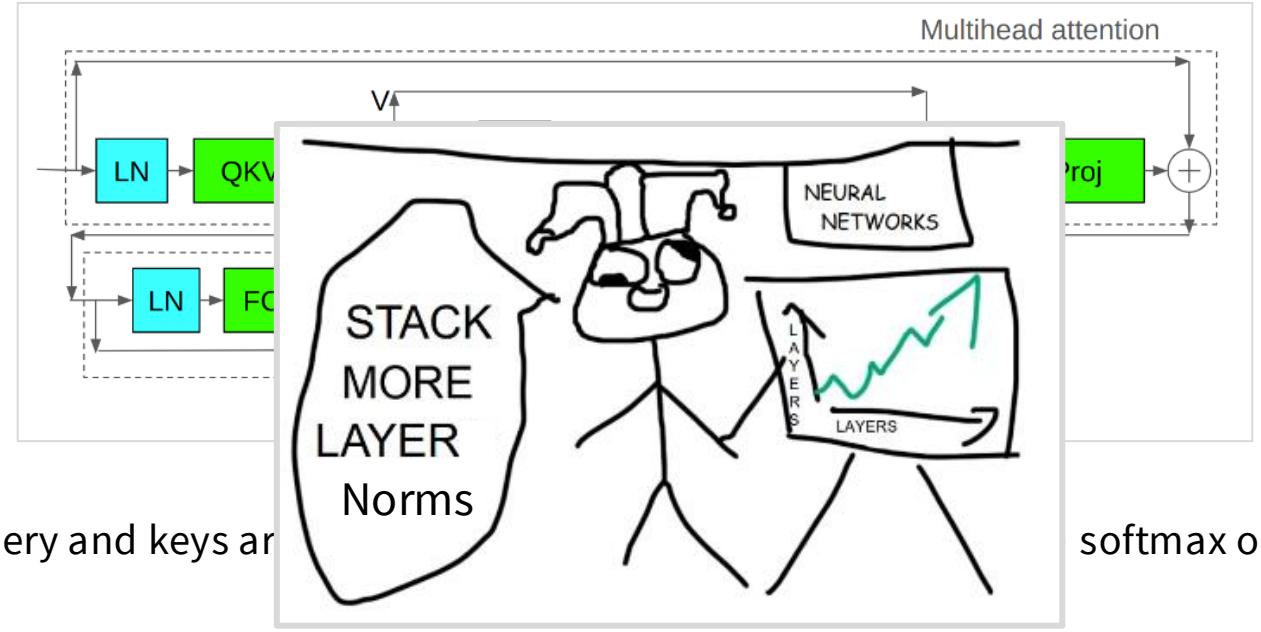
[From Devlin 2014]

This is useful for stability! PaLM used this ‘z loss’ trick.

> We additionally use an auxiliary loss of $z_loss = 10^{-4} \cdot \log^2 Z$ to encourage the softmax normalizer $\log(Z)$ to be close to 0, which we found increases the stability of training.

Other examples: Baichuan 2 (2023), DCLM (2024), OLMo 2 (2025), OLMo 3 (2025)

Attention softmax stability – the ‘QK norm’



The query and keys ar softmax operation.

Other examples: DCLM, OLMo2, Gemma 2, Qwen3, OLMo 3, Gemma 4

Originally from vision and multimodal models [Dehgani 2023, Idefcs, Chameleon]

Logit soft-capping.

Soft-capping the logits to some maximum value via Tanh

> **Logit soft-capping**. We cap logits (Bello et al., 2016) in each attention layer and the final layer such that the value of the logits stays between −soft_cap and +soft_cap. More specifically, we cap the logits with the following function:
>
> $$\text{logits} \leftarrow \text{soft_cap} * \tanh(\text{logits}/\text{soft_cap}).$$
>
> We set the soft_cap parameter to 50.0 for the self-attention layers and to 30.0 for the final layer.

Prevents logits from blowing up, but also might have perf issues?

Table 4: Models perplexity with confidence interval ±0.1 at 95% level.

bf16 baseline	*soft_cap*	*QKV_norm*	*QK_norm_cap*	*QK_norm*	*QK_FC_norm*
11.19	11.24	10.85	11.00	10.84	10.87

Attention heads

Most models don’t touch the attention heads much at all with a few minor exceptions..

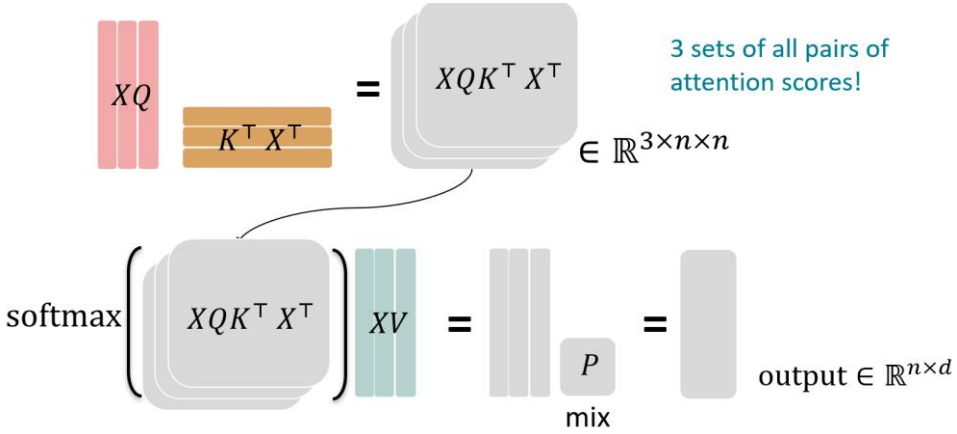
GQA / MQA : Saving inference costs by reducing the number of heads

Sparse or sliding window attention (GPT4/Mistral): restricting the attention pattern to reduce compute cost

Exotic SSM stuff (Jamba, Falcon 3, Qwen 3.5, etc): next lecture!

GQA/MQA – Reducing attention head cost

Let's think about the compute involved for attention



d = hidden dim
b = batch
n = length (<d)
h = heads
k = head dim (d/h)

X softmax projection

Total arithmetic operations (bnd^2), **total memory accesses** $(bnd + bhn^2 + d^2)$

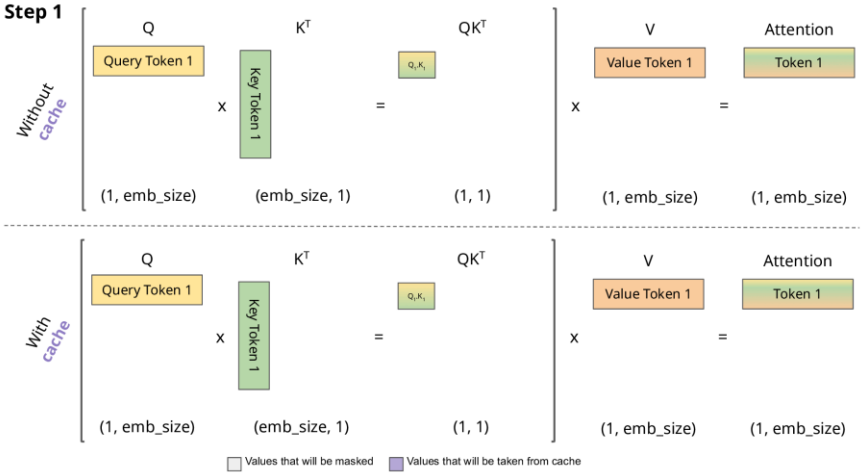
Arithmetic intensity (compute/memory) is high $O\left(\left(\frac{1}{k} + \frac{1}{bn}\right)^{-1}\right)$ - we can keep our GPUs running

GQA/MQA – Reducing attention head cost

What about the *incremental* case when we generate text?

Key difference: can't parallelize the generation process – needs to be step by step

In this case – we need to incrementally re-compute/update attention via the 'KV cache'



[Animation from https://medium.com/@joaolages/kv-caching-explained-276520203249]

GQA/MQA – Reducing attention head cost

What's the incremental arithmetic intensity?

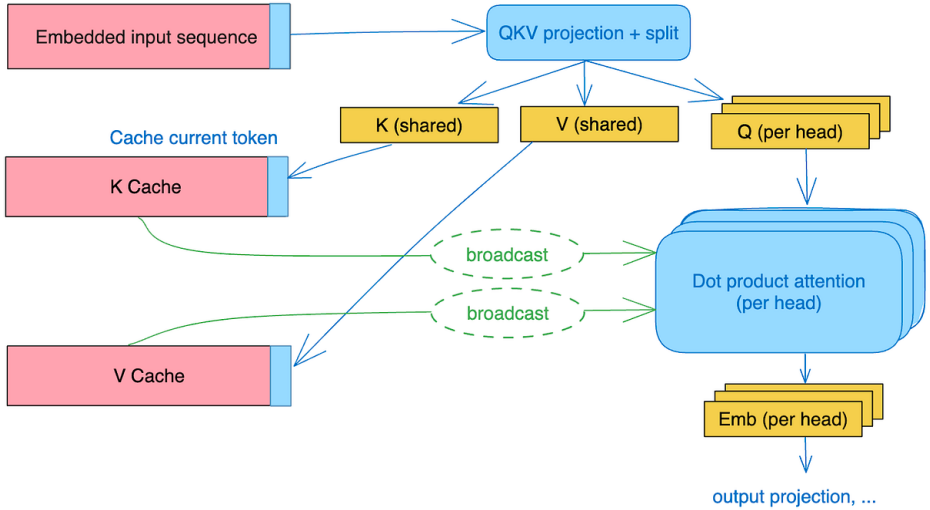
Total arithmetic operations (bnd^2), **total memory accesses** $(\overset{\text{K, V}}{bn^2d} + \overset{\text{projection}}{nd^2})$

Arithmetic intensity is not good $O\left(\left(\frac{n}{d} + \frac{1}{b}\right)^{-1}\right)$ - need large batches + short seq length (n) or big model dimensions (d)

Is there some way around this? The n/d term is difficult to reduce.

MQA – just have fewer key dimensions.

Key idea – have multiple queries, but just one dimension for keys and values



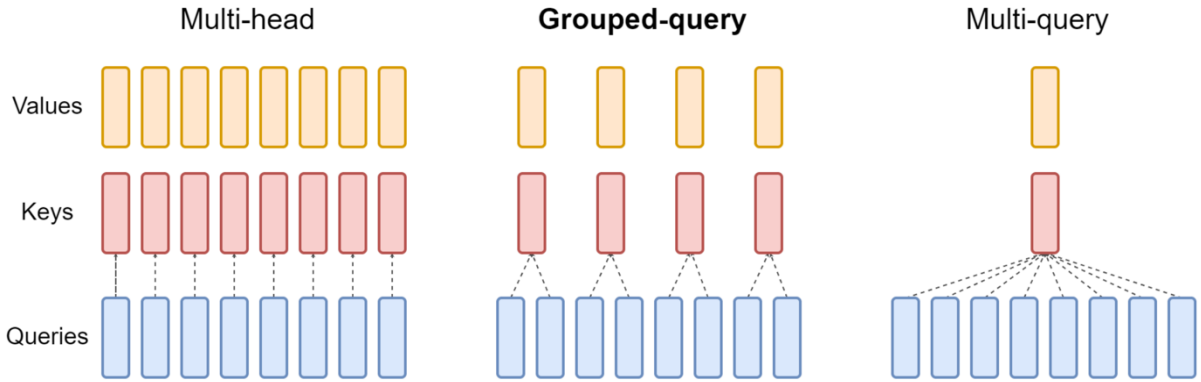
We have much fewer items to move in and out of memory (KV Cache)

Total memory access $(bnd + bn^2k + nd^2)$, **Arithmetic intensity** $O\left(\left(\frac{1}{d} + \frac{n}{dh} + \frac{1}{b}\right)^{-1}\right)$

[figure from https://blog.fireworks.ai/multi-query-attention-is-all-you-need-db072e758055]

Additional extensions – GQA

Don’t go all the way to one dimension of KV – have fewer dims



Simple knob to control expressiveness (key-query ratio) and inference efficiency

More recently – MLA (multihead latent attention) from deepseek v2

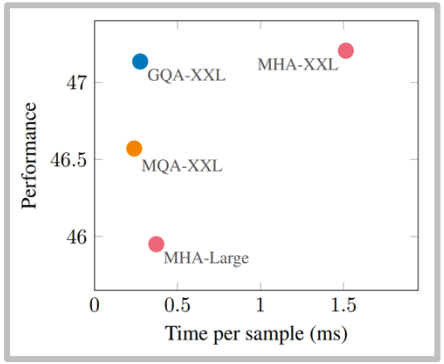
Does MQA hurt? Sometimes..

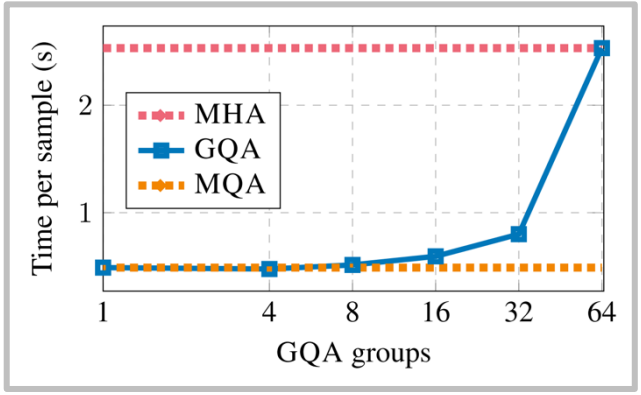
Small PPL hit w/ MQA [Shazeer 2019]

Table 3: Billion-Word LM Benchmark Results.

Attention	h	d_k, d_v	d_{ff}	dev-PPL
multi-head	8	128	8192	**29.9**
multi-query	8	128	9088	30.2
multi-head	1	128	9984	31.2
multi-head	2	64	9984	31.1
multi-head	4	32	9984	31.0
multi-head	8	16	9984	30.9

Low/no hit w/ GQA [Ainslie 2023]

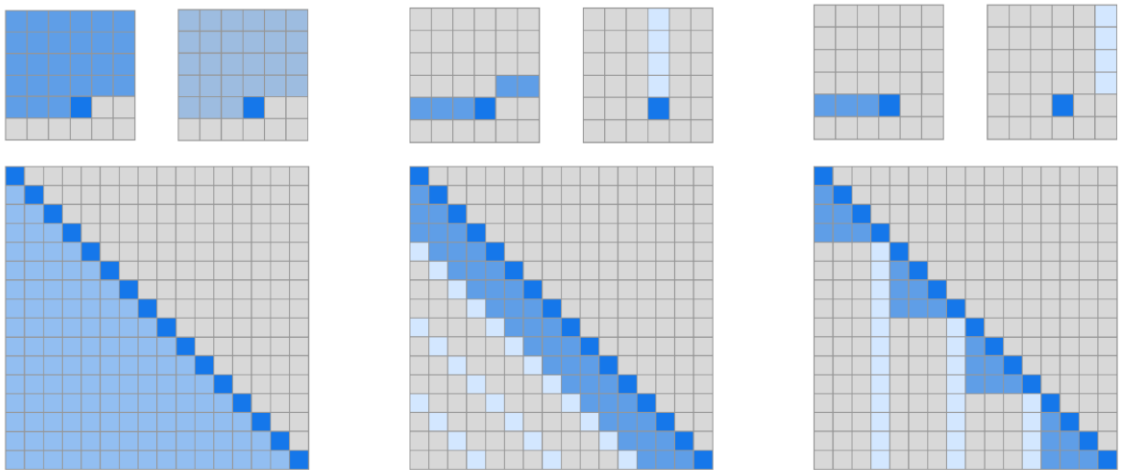




Sparse / sliding window attention

Attending to the entire context can be expensive (quadratic).

Build sparse / structured attention that trades off expressiveness vs runtime (GPT3, GPT-OSS, Gemma4)

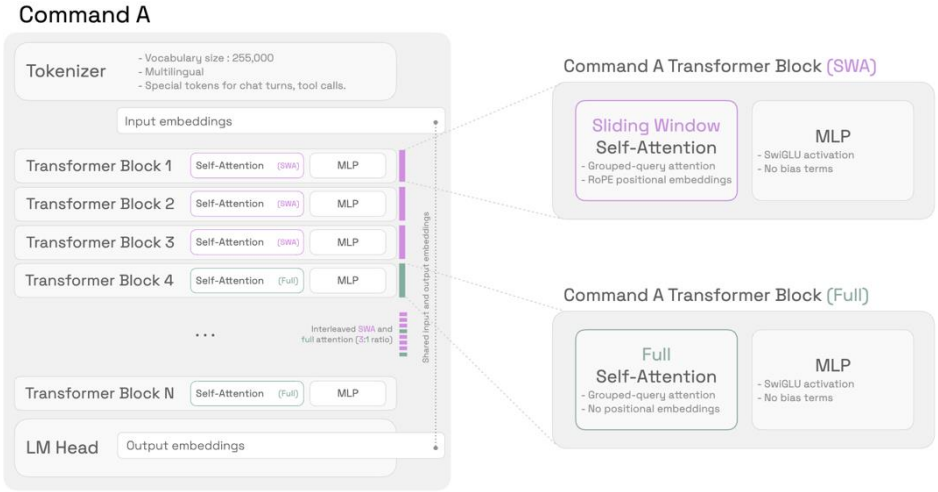


(a) Transformer (b) Sparse Transformer (strided) (c) Sparse Transformer (fixed)

[Child et al 2019]

Current standard trick – interleave ‘full’ and ‘LR’ attention

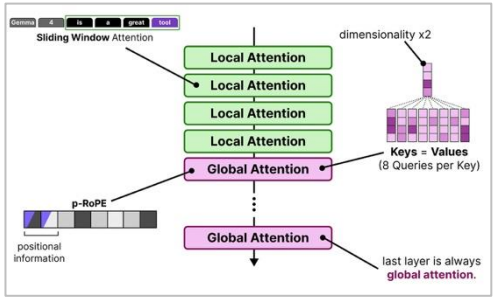
From Cohere Command A – Every 4th layer is a full attention



Long-range info via NoPE, short-range info via RoPE + SWA.

Other models – LLaMA 4, Gemma 3, Gemma 4, OLMo 3 does SWA+Full RoPE.

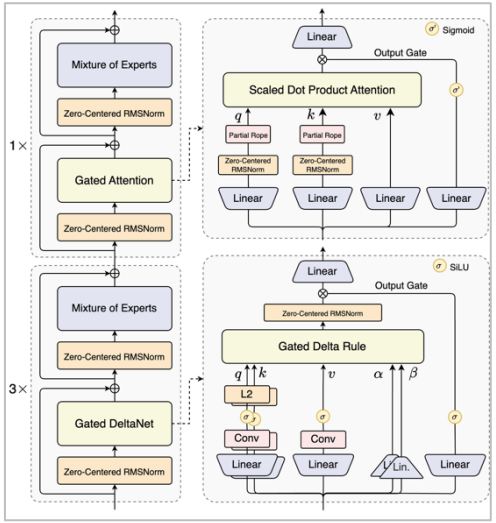
Other recent examples of interleaved attention



Gemma 4

Gradient clipping	1.0
Z-loss weight	10^{-5}
Weight decay on embeddings	No
Sliding window attention	3/4 of layers; 4,096 tokens
RoPE scaling	YaRN on full attn. layers
RoPE θ	$5 \cdot 10^5$
Layer norm applied to	Outputs

Olmo 3

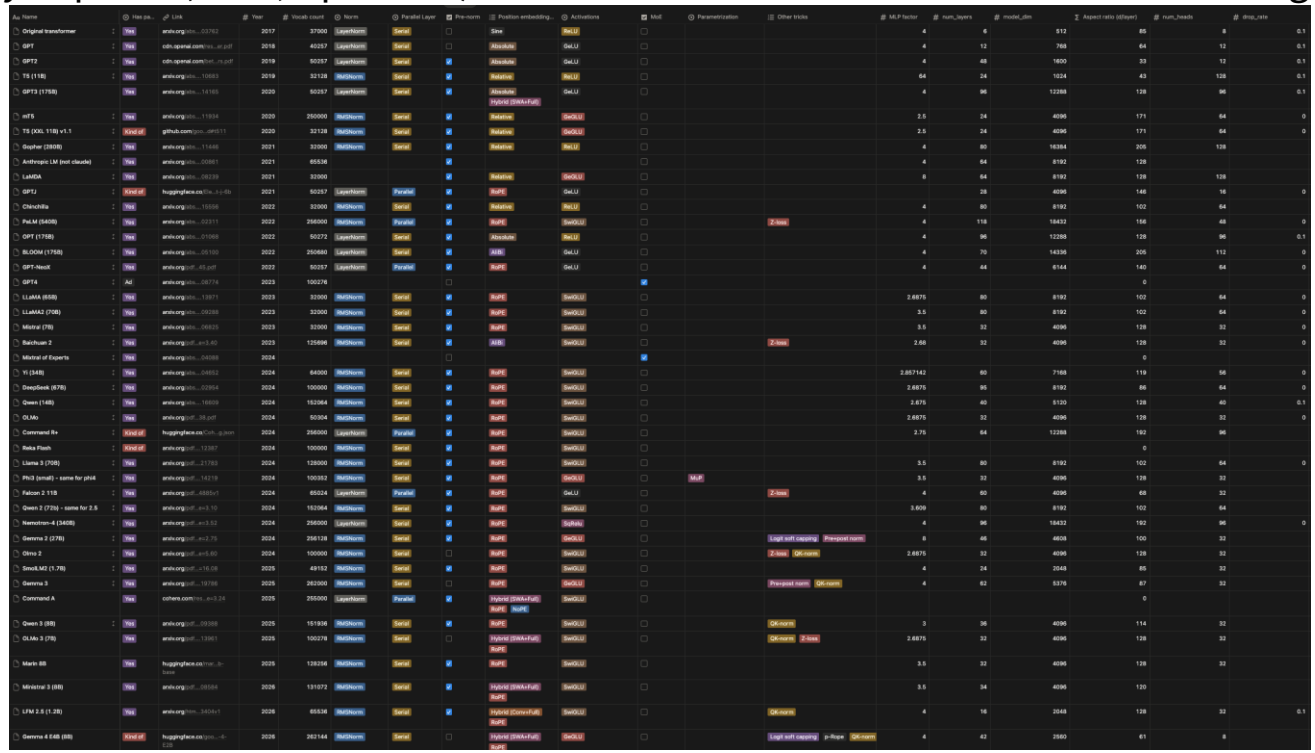


Qwen 3.5 / Qwen 3 Next

https://newsletter.maartengrootendorst.com/p/a-visual-guide-to-gemma-4

Recap, conclusion, etc.

Many aspects (arch, hparams) of transformers are in common across the big LMs



Major differences? Position embeddings, activations, tokenization